Satwik Paladugula

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GitHub | LinkedIn | Portfolio

Profile

Computer Science undergraduate with hands-on experience building AI-powered applications and backend systems using Python, Java, and SQL. Developed projects integrating Generative AI APIs with FastAPI and Spring Boot backends, demonstrating the ability to take ideas from concept to working software.

Education

B.Tech — Computer Science and Technology 2023–2027
Sri Vasavi Engineering College, Tadepalligudem CGPA: 8.25/10

Intermediate (PCM) 2021–2023
Bharatiya Vidya Bhavans 84.8%

SSC 2020–2021
Tirumala Proactive School 10/10 CGPA

Technical Skills

- **AI & Generative AI:** Gemini API, Large Language Models (LLMs), Prompt Engineering, NLP, Oracle Gen AI
- **Programming:** Python, Java, SQL
- **Backend Development:** FastAPI, Spring Boot, Spring Data JPA, Hibernate, REST APIs
- **Database:** MySQL, SQLite, DBMS
- **Enterprise Platforms:** ServiceNow (System Administration, Application Development), SAP ABAP Cloud
- **Tools & Concepts:** Git, GitHub, Postman, Streamlit, VS Code, OOP, API Design

Projects

ClarifySQL — AI-Powered Natural Language Data Assistant Python, FastAPI, Gemini API, MySQL

- Developed an AI assistant that converts plain English into schema-aware, validated SQL queries for MySQL databases using Google Gemini API.
- Implemented query understanding, clarification workflows, and SQL validation using SQLGlot for AST-level query safety.
- Built a FastAPI backend and Streamlit interface with result visualization, query history, and multi-table schema support.

CollectFast — B2B Invoicing & Payment Recovery Engine Python, FastAPI, UPI, ReportLab

- Built a zero-commission B2B invoicing SaaS generating dynamic UPI payment QR codes with WhatsApp-based payment recovery automation.
- Designed a FastAPI backend with RESTful endpoints for invoice management, payment tracking, and PDF generation using ReportLab.
- Deployed a live product demo on GitHub Pages demonstrating end-to-end invoicing with real-time payment link generation.

Student Management System Java, Spring Boot, MySQL

- Developed a RESTful backend using Spring Boot and MySQL for complete student record management with full CRUD operations.
- Implemented layered architecture using Spring Data JPA and Hibernate with clean separation of controller, service, and repository layers.
- Tested all API endpoints using Postman and maintained version control using Git and GitHub.

AI Security Gateway — LLM Reverse Proxy with Threat Detection Python, FastAPI, SQLite, Pydantic, Streamlit

- Built an enterprise reverse proxy between client applications and LLMs, intercepting and validating all requests and responses in real time.
- Implemented prompt injection detection and sensitive data leak prevention with sub-millisecond latency (0.14 ms p50).
- Designed a Streamlit monitoring dashboard for live threat visibility, request logging, and security policy management.

Certifications

- **Oracle Generative AI Professional** *Oracle*
- **ServiceNow Certified System Administrator (CSA)** *ServiceNow*
- **ServiceNow Certified Application Developer (CAD)** *ServiceNow*
- **SAP Certified Back-End Developer — ABAP Cloud** *SAP*
- **Salesforce Agentforce Specialist** *Salesforce*

Strengths

- **Problem Solving:** Strong analytical and logical thinking skills with the ability to break down complex problems into structured, working solutions
- **Quick Learner:** Independently acquired enterprise certifications in Oracle Gen AI, ServiceNow, SAP, and Salesforce
- **Communication:** Strong verbal and written communication; able to explain technical concepts clearly to diverse audiences